Reamy Womodowe Achirobe

MPhil Mechanical Engineering

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Research Interests

Machine learning for thermal-fluid systems |Energy storage systems |Multiphase flow and heat transfer | Physics-informed neural networks (PINNs) for renewable energy | Multiphase flow dynamics and heat transfer | Computational modeling of energy systems | Data-driven predictive maintenance | Thermal diagnostics and infrared thermography | Integration of AI with thermofluids engineering

Education

MPhil Mechanical Engineering **Expected:November 2025**

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Specialization: Thermofluids & Energy Systems

Thesis: Using Machine Learning and Infrared Thermography to Identify and Classify Defects among Crystalline Photovoltaics in Ghana

Advisors: Prof. Lena Dzifah Mensah & Prof. Muyiwa S. Adaramola

Key Coursework:

- Advanced Engineering Thermodynamics (81%) – Energy conversion, exergy analysis, thermodynamic cycles
- Heat and Mass Transfer (80%) – Conduction, convection, radiation, phase change phenomena
- Thermal Power Plant Engineering (83%) – Power cycles, CSP systems, component design
- Computational Fluid Dynamics (76%) – Numerical methods, turbulence modeling, validation
- Computer Aided Analysis and Modelling (79%) – ANSYS Fluent, thermal-fluid simulations
- Energy Conservation & Utilization (90%) – Renewable energy, sustainability, efficiency optimization
- Fluid Machinery (68%) – Pumps, turbines, compressors, cavitation analysis

BSc Mechanical Engineering **2016 – 2020**

Kwame Nkrumah University of Science and Technology(KNUST), Kumasi, Ghana

Specialization: Thermofluids & Energy Systems **Advisor:** Prof. David Ato Quansah

Research Experience

Graduate Researcher **September 2023 – Present**

KNUST,College of Engineering, Department of mechanical engineering, Kumasi, Ghana

MPhil Thesis: Using Machine Learning and Infrared Thermography to Identify and Classify Defects among Crystalline Photovoltaics in Ghana

Advisors: Prof. Lena Dzifah Mensah & Prof. Muyiwa S. Adaramola

- Developed deep learning framework using Convolutional Neural Networks (CNNs) in TensorFlow/Keras for automated thermal anomaly detection in PV modules, achieving high classification accuracy
- Designed experimental protocols for infrared thermography-based defect detection, combining thermal physics principles with computer vision techniques
- Implemented comprehensive image preprocessing pipeline including noise reduction, normalization, and data augmentation to enhance model robustness
- Integrated thermal imaging analysis with computational heat transfer modeling to predict defect propagation and performance degradation
- Conducted extensive literature review on physics-informed machine learning approaches for energy systems

- **Research Impact:** Work demonstrates potential for combining physics-based modeling with deep learning architectures , directly applicable to Physics-Informed Neural Networks (PINNs)
- **Skills Developed:** Thermal diagnostics, deep learning model training/validation, hyperparameter tuning, experimental design, scientific writing

DAAD Research Intern **September – November 2024**
Chair of Energy Process Engineering , University of Erlangen-Nuremberg, Bavaria, Germany
Project: Characterizing Fluidized Bed Bubbling Phenomena: Probing Dynamics with Dual-Sided Video and Pressure Analysis
Supervisor: Arkya Sanyal & Federica Torrigino

- Conducted experimental characterization of multiphase flow dynamics using high-speed video imaging and pressure transducers in bubbling fluidized bed reactors
- Applied advanced signal processing techniques (FFT-based spectral analysis) in Python (NumPy, SciPy, Pandas) to analyze pressure fluctuation data and identify flow regime transitions
- Developed predictive models for particle agglomeration using statistical analysis and machine learning approaches (scikit-learn)
- Performed thermal diagnostics using FIJI image processing software for flow visualization and regime identification
- Collaborated with international research team in English/German working environment
- **Research Impact:** Established methodology for early detection of the onset of agglomeration in fluidized beds used in biogenic fuel combustion.
- **Skills Developed:** Experimental thermal-fluids techniques, multiphase flow characterization, time-series analysis, international collaboration

Engineering Intern **August 2020 – August 2021**
Denys Engineers & Contractors, Accra, Ghana
Focus: Industrial Thermal-Fluid Systems

- Installed, maintained, and optimized centrifugal pumps, heat exchangers, and water distribution networks for industrial applications
- Conducted troubleshooting and root cause analysis for pump cavitation, fouling, and thermal system inefficiencies
- Gained practical understanding of real-world constraints in thermal system design, operation, and maintenance
- Developed hands-on expertise in HVAC systems, plumbing systems, and building mechanical infrastructure

Teaching Experience

Mathematics & Science Instructor **2021**
Light House Christian Mission School, Kumasi, Ghana

- Delivered STEM curriculum to junior high students, emphasizing problem-solving and critical thinking
- Developed assessments and interactive learning materials

Mathematics & Physics Tutor **2016 – 2020**
KANASU Vacation Classes, Kumasi, Ghana

- Conducted remedial courses and exam preparation for senior high students
- Delivered both in-person and online instruction using digital tools

Technical Skills

Programming & Computational Tools:

- **Python:** NumPy, Pandas, Matplotlib, SciPy, TensorFlow, Keras, PyTorch, scikit-learn – extensive experience in data analysis, machine learning pipelines, and thermal system modeling
- **MATLAB:** Numerical methods, data visualization, thermal-fluid system simulations

- **LaTeX:** Scientific document preparation, thesis writing
- **Git/GitHub:** Version control, collaborative development
- **SQL:** Database management, data querying

Deep Learning & Machine Learning:

- Convolutional Neural Networks (CNNs) for image classification and thermal anomaly detection
- Model training, validation, and hyperparameter optimization
- Transfer learning and model fine-tuning
- Experience with physics-informed machine learning architectures
- Statistical modeling and time-series analysis

Computational Fluid Dynamics & Thermal Analysis:

- **ANSYS Fluent:** Turbulence modeling (RANS, LES), multiphase flow simulations, heat transfer analysis
- Mesh generation, boundary condition specification, convergence analysis
- Post-processing and validation of CFD results
- **AutoCAD:** Technical drawing, system layout design

Experimental Techniques:

- Infrared thermography for thermal diagnostics
- High-speed imaging and flow visualization
- Pressure measurement systems and data acquisition
- FFT signal processing and spectral analysis
- Experimental design and uncertainty analysis

Data Science & Analysis:

- Data preprocessing, cleaning, and feature engineering
- Statistical analysis and hypothesis testing
- Time-series analysis for thermal and flow systems
- Data visualization (Matplotlib, Seaborn)

Domain Expertise:

- Advanced thermodynamics and heat transfer
- Multiphase flow dynamics and characterization
- Computational fluid dynamics and turbulence modeling
- Renewable energy systems (solar thermal, photovoltaics)
- Thermal power plant engineering and energy conversion
- Energy storage and efficiency optimization

Publications & Presentations

Thesis Research (In Progress):

- Achirobe Reamy Womodowe, Lena Dzifa Mensah, Muyiwa S. Adaramola ,"Using Machine Learning and Infrared Thermography to Identify and Classify Defects among Crystalline Photovoltaics in Ghana." *MPhil Thesis, KNUST.* **Expected submission: December 2025**.

Presentations:

- MPhil Viva for thesis presentation under: "Using Machine Learning and Infrared Thermography to Identify and Classify Defects among Crystalline Photovoltaics in Ghana." **Date: October,2025**

Honors & Awards

- **DAAD Research Internship Award**– Competitive German Academic Exchange Service scholarship for research at University of Erlangen-Nuremberg, Bavaria, Germany (**2024**)
- **KEEP Scholarship**– Full funding for MPhil studies at Kwame Nkrumah University of Science and Technology (**2023–2025**)

Professional Development & Certifications

- **Mastering the Art of Research Workshop** –: Research methodology, proposal writing, scientific communication, KNUST (**2025**)
- **Machine Learning Specialization(Course audit)** – Stanford University/Coursera: Supervised learning, neural networks, deep learning (**2023**)
- **Data Analytics with Python Nanodegree** – ALX Africa: Statistical analysis, data visualization (**2022**)
- **Google Data Analytics Professional Certificate**: Data cleaning, SQL, Tableau (**2021**)

Volunteering & Community Service

STEM Education Volunteer **2019 – 2023**
Ghana Education Service Community Outreach Program

- Volunteer mathematics and science instructor for underserved communities in Northern Ghana during school vacations
- Developed and delivered hands-on STEM workshops focusing on renewable energy and sustainable technology for over 200 students
- Created low-cost experimental demonstrations to teach thermodynamics and energy conversion principles
- Mentored high school students interested in pursuing engineering careers, providing guidance on university applications and career pathways

Engineering Mentor, KNUST Peer Mentorship Program **2018 – 2020**

- Provided academic mentorship to first and second-year mechanical engineering students
- Conducted study groups and tutoring sessions in thermodynamics, fluid mechanics, and heat transfer
- Assisted mentees with research project selection and experimental design

Community Solar Energy Awareness Campaign **2023 – 2024**

- Organized community workshops educating rural populations about solar energy technology and its applications
- Demonstrated solar panel installation and maintenance techniques to local technicians
- Collaborated with NGOs to promote renewable energy adoption in off-grid communities

Leadership & Service

Vice President, IAESTE KNUST Chapter **2024 – 2025**
International Association for the Exchange of Students for Technical Experience

- Coordinate international internship placements for engineering students across Africa and Europe
- Manage organizational operations, budgets, and partnerships with universities and companies
- Supervise team of 15+ student coordinators

Student Member, Professional Organizations **2025 – Present**

- Ghana Institution of Engineers (GhIE)
- National Society of Black Engineers (NSBE)

Organizer, KANASU KNUST Chapter **2019 – 2020**

- Coordinated academic and cultural activities for northern Ghanaian students
- Organized educational workshops and community outreach programs

Languages

- **English:** Fluent (academic and professional proficiency)
- **Twi:** Fluent (Ghanaian lingua franca)
- **Kasem:** Native (northern Ghanaian language)

Reference(s)

(available upon request)